



JEE Mains Formula Handbook

All Subjects

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PART I

Physics

1. Units & Measurements

Seven SI Base Units

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	kelvin	K
Amount of Substance	mole	mol
Luminous Intensity	candela	cd

Key Dimensional Formulas

Quantity	Dimensions
Velocity	$[LT^{-1}]$
Acceleration	$[LT^{-2}]$
Force	$[MLT^{-2}]$
Energy / Work / Torque	$[ML^2T^{-2}]$
Power	$[ML^2T^{-3}]$
Pressure / Stress	$[ML^{-1}T^{-2}]$
Momentum	$[MLT^{-1}]$
Angular Momentum	$[ML^2T^{-1}]$
G	$[M^{-1}L^3T^{-2}]$
Planck's h	$[ML^2T^{-1}]$
Surface tension	$[MT^{-2}]$
Viscosity η	$[ML^{-1}T^{-1}]$

Unit Conversion

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c \text{ for dimensions } [M^a L^b T^c]$$

Error Propagation

- Addition/subtraction: $\Delta Z = \Delta A + \Delta B$
- If $Z = A^p B^q / C^r$: $\frac{\Delta Z}{Z} = |p| \frac{\Delta A}{A} + |q| \frac{\Delta B}{B} + |r| \frac{\Delta C}{C}$
- Percentage error = $\frac{\Delta a}{a} \times 100\%$

Significant Figures

- Addition/subtraction: least decimal places
- Multiplication/division: least significant figures

Instruments

- Vernier LC = 1 MSD – 1 VSD = $\frac{1 \text{ mm}}{n}$
- Screw Gauge LC = $\frac{\text{Pitch}}{\text{No. of CSD}}$ (typically 0.01 mm)
- Reading = MSR + (VSR/CSR $\times$ LC) – Zero Error



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2. Kinematics

Equations of Motion (constant a)

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$
- $s = \frac{1}{2}(u + v)t$
- Distance in n th second: $s_n = u + \frac{a}{2}(2n - 1)$

Free Fall (from rest)

- $v = gt$; $h = \frac{1}{2}gt^2$; $v^2 = 2gh$
- Thrown up: max height $H = u^2/(2g)$, time of flight $T = 2u/g$

Projectile Motion (angle θ , speed u)

- $u_x = u \cos \theta$; $u_y = u \sin \theta$
- Time of flight: $T = \frac{2u \sin \theta}{g}$
- Max height: $H = \frac{u^2 \sin^2 \theta}{2g}$
- Range: $R = \frac{u^2 \sin 2\theta}{g}$; $R_{\max} = u^2/g$ at $\theta = 45^\circ$
- Trajectory: $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$
- Complementary angles θ and $(90^\circ - \theta)$ give same range

Horizontal Projectile (from height h)

- $t = \sqrt{2h/g}$; Range $= u\sqrt{2h/g}$; $v = \sqrt{u^2 + 2gh}$

Relative Velocity & Averages

- $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$
- Equal distances at v_1, v_2 : $v_{\text{avg}} = \frac{2v_1v_2}{v_1 + v_2}$ (harmonic)
- Equal times: $v_{\text{avg}} = (v_1 + v_2)/2$ (arithmetic)

Motion Graphs

- x - t : slope = velocity
- v - t : slope = acceleration; area = displacement
- a - t : area = Δv

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3. Laws of Motion**Newton's Laws & Momentum**

- $\vec{F}_{\text{net}} = m\vec{a}$; $\vec{F} = \frac{d\vec{p}}{dt}$; $\vec{p} = m\vec{v}$
- Impulse: $\vec{J} = \vec{F}\Delta t = \Delta\vec{p}$
- Third law: $\vec{F}_{AB} = -\vec{F}_{BA}$

Friction

- Static: $f_s \leq \mu_s N$; max: $f_{s,\text{max}} = \mu_s N$
- Kinetic: $f_k = \mu_k N$; $\mu_k < \mu_s$
- Incline: $N = mg \cos \theta$
- Down smooth incline: $a = g \sin \theta$
- Down rough incline: $a = g(\sin \theta - \mu_k \cos \theta)$
- Angle of repose: $\tan \theta_r = \mu_s$

Connected Systems

- Atwood: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$; $T = \frac{2m_1m_2g}{m_1 + m_2}$
- Table-pulley: $a = \frac{m_2g}{m_1 + m_2}$; $T = \frac{m_1m_2g}{m_1 + m_2}$

Circular Motion

- $a_c = v^2/r = \omega^2 r$; $F_c = mv^2/r$
- Banking (no friction): $\tan \theta = v^2/(rg)$
- Flat road max speed: $v_{\max} = \sqrt{\mu_s r g}$



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4. Work, Energy & Power

Work & Energy

- $W = Fd \cos \theta = \vec{F} \cdot \vec{d}$; Variable: $W = \int F dx$
- $KE = \frac{1}{2}mv^2$; $PE_{\text{grav}} = mgh$; $PE_{\text{spring}} = \frac{1}{2}kx^2$
- Work-Energy theorem: $W_{\text{net}} = \Delta KE$
- Conservation: $KE_1 + PE_1 + W_{\text{nc}} = KE_2 + PE_2$
- Spring work (natural to x): $W_{\text{ext}} = \frac{1}{2}kx^2$

Power

- $P_{\text{avg}} = W/t$; $P = \vec{F} \cdot \vec{v} = Fv \cos \theta$
- At max speed: $P = F_{\text{resistance}} \times v_{\max}$
- 1 hp = 746 W; 1 kWh = 3.6×10^6 J

Collisions

- Momentum conserved always: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$
- Coefficient of restitution: $e = \frac{v_2 - v_1}{u_1 - u_2}$
- Elastic ($e = 1$): KE conserved
- Perfectly inelastic ($e = 0$): $v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$
- KE loss (perfectly inelastic): $\Delta KE = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2$
- Elastic ($u_2 = 0$): $v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1$; $v_2 = \frac{2m_1}{m_1 + m_2} u_1$
- Bounce height: $h' = e^2 h$; After n bounces: $h_n = e^{2n} h$

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5. Rotational Motion

Rotational Kinematics (constant α)

- $\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; $\omega^2 = \omega_0^2 + 2\alpha\theta$
- $v = r\omega$; $a_t = r\alpha$; $a_c = r\omega^2 = v^2/r$
- rpm to rad/s: $\omega = 2\pi N/60$

Moments of Inertia

Body (axis)	I
Ring / Hollow cyl. (central axis)	MR^2
Ring (diameter)	$\frac{1}{2}MR^2$
Disc / Solid cyl. (central axis)	$\frac{1}{2}MR^2$
Disc (diameter)	$\frac{1}{4}MR^2$
Solid sphere (diameter)	$\frac{2}{5}MR^2$
Hollow sphere (diameter)	$\frac{2}{3}MR^2$
Rod (centre, $\perp$)	$\frac{1}{12}ML^2$
Rod (end, $\perp$)	$\frac{1}{3}ML^2$

Theorems

- Parallel axis: $I = I_{cm} + Md^2$
- Perpendicular axis (2D only): $I_z = I_x + I_y$

Dynamics

- Torque: $\tau = rF \sin \theta$; $\tau_{net} = I\alpha$
- Angular momentum: $L = I\omega$; $\tau = dL/dt$
- Conservation: if $\tau_{ext} = 0$, $I_1\omega_1 = I_2\omega_2$
- $KE_{rot} = \frac{1}{2}I\omega^2$; $W = \tau\theta$; $P = \tau\omega$

Rolling Without Slipping

- Constraint: $v_{cm} = R\omega$; $a_{cm} = R\alpha$
- Total KE = $\frac{1}{2}Mv^2(1 + k)$ where $k = I_{cm}/(MR^2)$
- Acceleration down incline: $a = \frac{g \sin \theta}{1 + k}$
- Speed at bottom: $v = \sqrt{2gh/(1 + k)}$
- Order (fastest first): solid sphere, disc, hollow sphere, ring



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6. Gravitation

Newton's Law & g

- $F = Gm_1m_2/r^2$; $G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
- $g = GM/R^2$; $g \approx 9.8 \text{ m/s}^2$
- Height: $g_h = g/(1 + h/R)^2$; approx: $g(1 - 2h/R)$
- Depth: $g_d = g(1 - d/R)$; at centre: $g = 0$
- Latitude: $g_\lambda = g - R\omega^2 \cos^2 \lambda$

Potential & Energy

- Field: $E_g = GM/r^2$; Potential: $V = -GM/r$
- PE: $U = -Gm_1m_2/r$; (near surface: $U \approx mgh$)
- Binding energy (surface): $BE = GMm/R = mgR$

Satellites & Escape

- Orbital velocity: $v_o = \sqrt{GM/r} = \sqrt{gR^2/(R+h)}$
- Escape velocity: $v_e = \sqrt{2gR} = \sqrt{2} v_o$ (at surface)
- Time period: $T = 2\pi\sqrt{r^3/(GM)}$
- Energy: $E = -GMm/(2r)$; $KE = -E$; $PE = 2E$
- Geostationary: $T = 24 \text{ hr}$, $h \approx 36000 \text{ km}$
- Kepler III: $T^2 \propto a^3$

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7. Properties of Solids & Liquids

Elasticity

- Stress = F/A ; Strain = $\Delta L/L$ (or $\Delta V/V$, ϕ)
- Hooke: Stress = $E \times$ Strain (within proportional limit)
- Young's: $Y = FL/(A\Delta L)$
- Bulk: $B = -V\Delta P/\Delta V$; Compressibility = $1/B$
- Shear: $G = (F/A)/\phi$
- Poisson's ratio: $\sigma = -(\Delta d/d)/(\Delta L/L)$
- Elastic PE per unit volume: $u = \frac{1}{2} \times \text{stress} \times \text{strain}$
- Total PE: $U = \frac{1}{2} F \Delta L$

Fluid Statics

- Hydrostatic pressure: $P = P_0 + \rho gh$
- Pascal's law: $F_1/A_1 = F_2/A_2$
- Buoyancy: $F_b = \rho_f V_{\text{sub}} g$
- Floating: $V_{\text{sub}}/V = \rho_{\text{body}}/\rho_{\text{fluid}}$
- Apparent weight: $W_{\text{app}} = W - F_b$

Fluid Dynamics

- Continuity: $A_1 v_1 = A_2 v_2$
- Bernoulli: $P + \frac{1}{2} \rho v^2 + \rho gh = \text{const}$
- Torricelli (efflux): $v = \sqrt{2gh}$
- Venturi: $P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2)$

Viscosity

- Newton: $F = \eta A dv/dy$
- Stokes: $F = 6\pi\eta r v$
- Terminal velocity: $v_t = \frac{2r^2(\rho_s - \rho_f)g}{9\eta}$
- Poiseuille: $Q = \pi \Delta P r^4 / (8\eta L)$
- Reynolds: $Re = \rho v D / \eta$; laminar if $Re < 2000$

Surface Tension

- $S = F/l$; Surface energy = $S \times \Delta A$
- Drop: $\Delta P = 2S/R$; Soap bubble: $\Delta P = 4S/R$
- Capillary rise: $h = 2S \cos \theta / (\rho g r)$
- Angle of contact: water-glass $\approx 0^\circ$; mercury-glass $\approx 137^\circ$



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8. Thermodynamics & Kinetic Theory

Ideal Gas & First Law

- $PV = nRT = Nk_B T$; $R = 8.314 \text{ J/mol K}$
- $k_B = 1.38 \times 10^{-23} \text{ J/K}$; $N_A = 6.022 \times 10^{23}$
- First law: $Q = \Delta U + W$; $\Delta U = nC_v \Delta T$ (always for ideal gas)
- $W_{\text{gas}} = \int P dV$; PV diagram area = work

Thermodynamic Processes

Process	Const.	W	ΔU	Q
Isothermal	T	$nRT \ln \frac{V_2}{V_1}$	0	W
Adiabatic	$Q=0$	$\frac{nR(T_1-T_2)}{\gamma-1}$	$-W$	0
Isobaric	P	$P\Delta V$	$nC_v\Delta T$	$nC_p\Delta T$
Isochoric	V	0	$nC_v\Delta T$	ΔU

Adiabatic Relations

- $PV^\gamma = \text{const}$; $TV^{\gamma-1} = \text{const}$; $T^\gamma P^{1-\gamma} = \text{const}$

Carnot Engine & Entropy

- Efficiency: $\eta = 1 - T_C/T_H = W/Q_H$
- COP (refrigerator): $\beta = Q_C/W = T_C/(T_H - T_C)$
- Entropy change: $\Delta S = \int dQ_{\text{rev}}/T$
- Cyclic process: $\Delta U = 0$; $Q_{\text{net}} = W_{\text{net}}$

Kinetic Theory

- Pressure: $P = \frac{1}{3}\rho v_{\text{rms}}^2$
- KE per molecule: $\frac{3}{2}k_B T$; per mole: $\frac{3}{2}RT$
- $v_{\text{rms}} = \sqrt{3RT/M}$; $v_{\text{avg}} = \sqrt{8RT/(\pi M)}$; $v_{\text{mp}} = \sqrt{2RT/M}$
- Ratio: $v_{\text{mp}} : v_{\text{avg}} : v_{\text{rms}} = \sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$

Degrees of Freedom & Specific Heats

Gas	f	C_v	γ
Monoatomic	3	$\frac{3}{2}R$	5/3
Diatomic (rigid)	5	$\frac{5}{2}R$	7/5
Polyatomic	6+	$\frac{f}{2}R$	$(f+2)/f$
Mayer: $C_p - C_v = R$; $\gamma = 1 + 2/f$			

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9. Oscillations

SHM Equations

- Condition: $a = -\omega^2 x$; $F = -kx$
- $x = A \sin(\omega t + \phi)$
- $v = A\omega \cos(\omega t + \phi) = \omega \sqrt{A^2 - x^2}$
- $a = -A\omega^2 \sin(\omega t + \phi) = -\omega^2 x$
- $v_{\max} = A\omega$ (at $x = 0$); $a_{\max} = A\omega^2$ (at $x = \pm A$)

Time Period & Energy

- $T = 2\pi/\omega = 2\pi\sqrt{m/k}$; $\omega = \sqrt{k/m}$
- Total energy: $E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2$ (constant)
- $KE = \frac{1}{2}k(A^2 - x^2)$; $PE = \frac{1}{2}kx^2$
- $KE = PE$ at $x = A/\sqrt{2}$
- KE and PE oscillate at frequency 2ω

Pendulums & Springs

- Simple pendulum: $T = 2\pi\sqrt{l/g}$
- Compound: $T = 2\pi\sqrt{I/(mgd)}$
- In lift (up, acc. a): $g_{\text{eff}} = g + a$; (down): $g_{\text{eff}} = g - a$
- Free fall: $T \rightarrow \infty$; Horizontal acc. a : $g_{\text{eff}} = \sqrt{g^2 + a^2}$
- Springs parallel: $k_{\text{eff}} = k_1 + k_2$
- Springs series: $1/k_{\text{eff}} = 1/k_1 + 1/k_2$
- Superposition: $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$

Damped & Forced

- Damped: $x = A_0 e^{-bt/(2m)} \sin(\omega' t + \phi)$; $\omega' = \sqrt{\omega_0^2 - (b/2m)^2}$
- Resonance: max amplitude when $\omega_d \approx \omega_0$

Tip: If three points are collinear, the cross product of any two displacement vectors is zero.



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10. Waves

Wave Fundamentals

- $v = f\lambda = \omega/k$; $k = 2\pi/\lambda$; $\omega = 2\pi f$
- Progressive: $y = A \sin(kx - \omega t)$ (+x direction)
- String: $v = \sqrt{T/\mu}$; $\mu = m/L$
- Sound in gas: $v = \sqrt{\gamma RT/M}$; $v \propto \sqrt{T}$
- Sound in solid: $v = \sqrt{Y/\rho}$; In liquid: $v = \sqrt{B/\rho}$

Standing Waves & Harmonics

- Standing: $y = 2A \sin(kx) \cos(\omega t)$
- Node spacing = $\lambda/2$; Node to antinode = $\lambda/4$
- String (both ends fixed) / Open pipe: $f_n = nv/(2L)$, all n
- Closed pipe: $f_n = nv/(4L)$, odd n only
- Fundamental (closed) = $\frac{1}{2} \times$ fundamental (open) for same L

Beats & Doppler

- Beats: $f_{\text{beat}} = |f_1 - f_2|$
- Doppler: $f' = f_0 \left(\frac{v \pm v_o}{v \mp v_s} \right)$
- "Toward" $\rightarrow$ higher f : add v_o in numerator, subtract v_s in denominator
- Intensity: $I = P/(4\pi r^2)$; $\beta = 10 \log(I/I_0)$ dB

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11. Electrostatics**Coulomb's Law & Fields**

- $F = kq_1q_2/r^2$; $k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$
- $e = 1.6 \times 10^{-19} \text{ C}$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
- Point charge: $E = kq/r^2$; $V = kq/r$
- In medium: $F_{\text{med}} = F_{\text{vac}}/\kappa$

Field Formulas

- Infinite line (λ/m): $E = \lambda/(2\pi\epsilon_0 r)$
- Infinite sheet (σ): $E = \sigma/(2\epsilon_0)$
- Two parallel plates: $E_{\text{between}} = \sigma/\epsilon_0$
- Ring (axis, distance x): $E = kQx/(R^2 + x^2)^{3/2}$
- Charged sphere (outside): $E = kQ/r^2$; (inside conductor): $E = 0$

- Non-conducting sphere (inside): $E = kQr/R^3$

Electric Dipole

- $\vec{p} = q(2l)\hat{d}$ (negative to positive)
- Axial field: $E = 2kp/r^3$; Equatorial: $E = kp/r^3$
- In uniform field: $\tau = pE \sin \theta$; $U = -pE \cos \theta$
- In non-uniform field: $F = p dE/dx$

Potential & Energy

- $V = kq/r$; $\vec{E} = -dV/dr$
- Potential energy: $U = kq_1q_2/r$
- System of n charges: $\frac{n(n-1)}{2}$ pairs
- Equipotential surface $\perp$ field lines; no work along it

Gauss's Law

$$\oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}/\epsilon_0$$

Capacitors

- $C = Q/V$; Parallel plate: $C = \epsilon_0 A/d$
- With dielectric: $C = \kappa \epsilon_0 A/d$
- Parallel: $C_{\text{eq}} = C_1 + C_2$; Series: $1/C_{\text{eq}} = 1/C_1 + 1/C_2$
- Energy: $U = \frac{1}{2} CV^2 = Q^2/(2C) = \frac{1}{2} QV$
- Energy density: $u = \frac{1}{2} \epsilon_0 E^2$
- Partial dielectric: $C = \epsilon_0 A/(d - t + t/\kappa)$



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12. Current Electricity

Ohm's Law & Resistance

- $V = IR$; $R = \rho l/A$; $J = I/A = \sigma E$
- $I = nAv_d e$; $v_d = eE\tau/m$
- Temperature: $R_T = R_0(1 + \alpha \Delta T)$
- Series: $R_{\text{eq}} = R_1 + R_2 + \dots$
- Parallel: $1/R_{\text{eq}} = 1/R_1 + 1/R_2 + \dots$
- Two in parallel: $R_{\text{eq}} = R_1 R_2 / (R_1 + R_2)$

Kirchhoff & Circuits

- KCL: $\sum I_{\text{in}} = \sum I_{\text{out}}$
- KVL: $\sum \Delta V = 0$ around any loop
- Wheatstone: $P/Q = R/S$ (balanced, no galvanometer current)
- Meter bridge: $R/S = l/(100 - l)$
- Potentiometer: $\mathcal{E}_1/\mathcal{E}_2 = l_1/l_2$

Cells & Power

- Terminal voltage: $V = \mathcal{E} - Ir$
- Series: $\mathcal{E}_{\text{tot}} = n\mathcal{E}$, $r_{\text{tot}} = nr$
- Parallel: $\mathcal{E}_{\text{tot}} = \mathcal{E}$, $r_{\text{tot}} = r/m$
- Power: $P = VI = I^2R = V^2/R$
- Heat: $H = I^2Rt$
- Max power transfer: $R = r$; $P_{\text{max}} = \mathcal{E}^2/(4r)$

Galvanometer Conversion

- To ammeter (shunt S in parallel): $S = I_g G / (I - I_g)$
- To voltmeter (resistance R in series): $R = V/I_g - G$

Tip: “At least one” problems: use $P(\text{at least } 1) = 1 - P(\text{none})$.

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13. Magnetic Effects & Magnetism**Magnetic Field Sources**

- Biot-Savart: $dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$; $\mu_0 = 4\pi \times 10^{-7}$
- Infinite wire: $B = \mu_0 I / (2\pi d)$
- Finite wire: $B = \frac{\mu_0 I}{4\pi d} (\sin \phi_1 + \sin \phi_2)$
- Circular loop centre: $B = \mu_0 NI / (2R)$
- Loop axis: $B = \mu_0 I R^2 / [2(R^2 + x^2)^{3/2}]$
- Arc (θ rad): $B = \mu_0 I \theta / (4\pi R)$
- Solenoid: $B = \mu_0 nI$; At end: $B = \mu_0 nI / 2$
- Toroid: $B = \mu_0 NI / (2\pi r)$

Ampere's Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

Forces

- Lorentz: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$
- Magnetic: $F = qvB \sin \theta$; No work done by $\vec{F}_B$
- Circular motion: $r = mv/(qB)$; $T = 2\pi m/(qB)$; $f = qB/(2\pi m)$
- Helical pitch: $p = v_{\parallel} T = 2\pi mv \cos \theta / (qB)$
- Wire: $F = BIl \sin \theta$
- Parallel wires: $F/l = \mu_0 I_1 I_2 / (2\pi d)$ (attract if same direction)

Magnetic Dipole & Magnetism

- $m = NIA$; $\tau = mB \sin \theta$; $U = -mB \cos \theta$
- Bar magnet: axial $B = \mu_0 2M / (4\pi r^3)$; equatorial $B = \mu_0 M / (4\pi r^3)$
- Earth: $B_H = B \cos \phi$; $B_V = B \sin \phi$; $\tan \phi = B_V / B_H$
- Magnetic moment: $\mu = \sqrt{n(n+2)} \text{ BM}$
- Dia: $\chi < 0$; Para: $\chi > 0$ small; Ferro: $\chi \gg 0$
- Curie law: $\chi \propto 1/T$ (para); Curie temp. T_C (ferro $\rightarrow$ para)



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14. Electromagnetic Induction & AC

EMI Formulas

- Flux: $\phi_B = BA \cos \theta$ (Wb)
- Faraday: $\mathcal{E} = -Nd\phi_B/dt$; Lenz: opposes change
- Motional (rod): $\mathcal{E} = Blv$
- Rotating rod: $\mathcal{E} = \frac{1}{2} Bl^2 \omega$
- Rotating coil: $\mathcal{E} = NBA\omega \sin(\omega t)$

Inductance

- Self: $\mathcal{E} = -LdI/dt$; $L = \mu_0 N^2 A / l$ (solenoid)
- Mutual: $\mathcal{E}_2 = -MdI_1/dt$; $M = k\sqrt{L_1 L_2}$
- Energy: $U = \frac{1}{2} LI^2$; Density: $u = B^2 / (2\mu_0)$
- Inductors series: $L = L_1 + L_2$; Parallel: $1/L = 1/L_1 + 1/L_2$

AC Circuits

- $V_{\text{rms}} = V_0 / \sqrt{2}$; $V_{\text{avg}} = 2V_0 / \pi$
- AC generator peak: $\mathcal{E}_0 = NBA\omega$

- $X_L = \omega L$; $X_C = 1/(\omega C)$
- R only: V, I in phase
- L only: I lags V by 90°
- C only: I leads V by 90° (CIVIL mnemonic)

Series LCR

- $Z = \sqrt{R^2 + (X_L - X_C)^2}$; $I = V/Z$
- Phase: $\tan \phi = (X_L - X_C)/R$
- Resonance: $\omega_0 = 1/\sqrt{LC}$; $f_0 = 1/(2\pi\sqrt{LC})$
- At resonance: $Z = R$, I max, $\phi = 0$
- Q-factor: $Q = \omega_0 L/R = (1/R)\sqrt{L/C}$
- $V_L = V_C = QV$ at resonance

Power & Transformer

- Average power: $P = V_{\text{rms}} I_{\text{rms}} \cos \phi = I^2 R$
- Power factor: $\cos \phi = R/Z$
- Pure L or C: $\cos \phi = 0$, wattless current
- Transformer: $V_s/V_p = N_s/N_p = I_p/I_s$
- Efficiency: $\eta = V_s I_s / (V_p I_p) \times 100\%$

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15. Electromagnetic Waves

Key Formulas

- $c = 1/\sqrt{\mu_0 \epsilon_0} = 3 \times 10^8$ m/s
- In medium: $v = c/n$; $n = \sqrt{\mu_r \epsilon_r}$
- $E_0/B_0 = c$; E and B in phase, $\perp$ to each other and to propagation
- Intensity: $I = \frac{1}{2} \epsilon_0 c E_0^2 = c B_0^2 / (2\mu_0)$
- Energy density: $\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = B_0^2 / (2\mu_0)$
- Radiation pressure: absorbing $P = I/c$; reflecting $P = 2I/c$
- Displacement current: $I_d = \epsilon_0 d\Phi_E/dt$

EM Spectrum (increasing frequency)

Radio → Microwave → IR → Visible → UV → X-ray → γ -ray

Visible: Red (700 nm) → Violet (400 nm)

Tip: In coordinate geometry, check for degenerate conics (pair of lines) before solving.



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16. Optics

Reflection — Mirrors

- $i = r$; $f = R/2$
- Mirror formula: $1/v + 1/u = 1/f$; $m = -v/u$
- Concave: $f < 0$; Convex: $f > 0$
- Plane mirror: image at same distance, virtual, laterally inverted

Refraction

- Snell: $n_1 \sin \theta_1 = n_2 \sin \theta_2$; $n = c/v$
- Critical angle (TIR): $\sin \theta_c = n_2/n_1 = 1/n$ (to air)
- TIR: denser $\rightarrow$ rarer, $\theta > \theta_c$
- Spherical surface: $n_2/v - n_1/u = (n_2 - n_1)/R$

Thin Lenses

- $1/v - 1/u = 1/f$; $m = v/u$; $P = 1/f$ (diopetre, f in m)
- Lensmaker: $1/f = (n - 1)(1/R_1 - 1/R_2)$
- Lenses in contact: $1/f = 1/f_1 + 1/f_2$; $P = P_1 + P_2$
- Separated by d : $1/f = 1/f_1 + 1/f_2 - d/(f_1 f_2)$

Prism

- $\delta = (i_1 + i_2) - A$; $r_1 + r_2 = A$
- Min deviation: $i_1 = i_2$; $n = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(A/2)}$
- Thin prism: $\delta = (n - 1)A$
- Dispersion: $\theta = (n_V - n_R)A$; $\omega = (n_V - n_R)/(n_Y - 1)$

Optical Instruments

- Simple microscope: $M = 1 + D/f$ (near point); $M = D/f$ (infinity)
- Compound microscope: $M = -(L/f_o)(D/f_e)$ (at infinity)
- Telescope: $M = -f_o/f_e$; Tube length = $f_o + f_e$
- $D = 25$ cm (least distance of distinct vision)

Wave Optics

- YDSE fringe width: $\beta = \lambda D/d$
- Bright: $y_n = n\lambda D/d$; Dark: $y_n = (2n+1)\lambda D/(2d)$
- Path difference: $\Delta = yd/D$
- Single slit: central max width = $2\lambda D/a$; minima: $a \sin \theta = n\lambda$
- Malus: $I = I_0 \cos^2 \theta$; Unpolarized through polarizer: $I_0/2$
- Brewster: $\tan \theta_B = n$
- In medium: $\lambda' = \lambda/n$; $\beta' = \beta/n$

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17. Dual Nature of Matter & Radiation

Photons

- Energy: $E = h\nu = hc/\lambda$; $h = 6.626 \times 10^{-34} \text{ J s}$
- Momentum: $p = h/\lambda = E/c$
- $E \text{ (eV)} = 1240/\lambda \text{ (nm)} = 12400/\lambda \text{ (\AA)}$
- Rest mass of photon = 0

Photoelectric Effect

- Einstein: $K_{\max} = h\nu - \phi = h(\nu - \nu_0)$
- Stopping potential: $eV_0 = h\nu - \phi$
- Threshold: $\nu_0 = \phi/h$; $\lambda_0 = hc/\phi$
- $K_{\max}$, V_0 depend on frequency, not intensity
- Photocurrent depends on intensity (above ν_0)

de Broglie Wavelength

- $\lambda = h/p = h/(mv) = h/\sqrt{2mK}$
- Electron through V volts: $\lambda = 12.27/\sqrt{V} \text{ \AA}$
- Thermal: $\lambda = h/\sqrt{3mk_B T}$
- Lighter particle at same KE $\rightarrow$ longer λ



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18. Atoms & Nuclei

Bohr Model (H-like, charge Ze)

- $r_n = 0.529 n^2 / Z \text{ \AA}$; $v_n = 2.18 \times 10^6 Z / n \text{ m/s}$
- $E_n = -13.6 Z^2 / n^2 \text{ eV}$; $KE_n = -E_n$; $PE_n = 2E_n$
- Ionization energy of H = 13.6 eV (from $n = 1$)
- $1/\lambda = RZ^2(1/n_f^2 - 1/n_i^2)$; $R = 1.097 \times 10^7 \text{ m}^{-1}$
- Spectral lines from n : $n(n-1)/2$

Spectral Series

Series	n_f	n_i	Region
Lyman	1	2, 3, ...	UV
Balmer	2	3, 4, ...	Visible
Paschen	3	4, 5, ...	IR
Brackett	4	5, 6, ...	IR
Pfund	5	6, 7, ...	Far IR

X-Rays

- $\lambda_{\min} = hc/(eV) = 12400/V \text{ \AA}$
- Moseley: $\sqrt{\nu} = a(Z - b)$; $b \approx 1$ for K-series

Nuclear Physics

- $R = R_0 A^{1/3}$; $R_0 \approx 1.2 \text{ fm}$; Nuclear density $\approx 2.3 \times 10^{17} \text{ kg/m}^3$
- Mass defect: $\Delta m = Zm_p + Nm_n - M_{\text{nuc}}$
- $BE = \Delta m \times 931.5 \text{ MeV}$; BE/A peaks at Fe-56
- $1 \text{ u} = 931.5 \text{ MeV/c}^2$

Radioactivity

- α : $A \rightarrow A-4$, $Z \rightarrow Z-2$; β^- : $Z \rightarrow Z+1$; γ : no change in A, Z
- $N = N_0 e^{-\lambda t}$; $A = \lambda N = A_0 e^{-\lambda t}$
- $t_{1/2} = 0.693/\lambda$; $\tau = 1/\lambda = 1.44 t_{1/2}$
- After n half-lives: $N = N_0/2^n$
- Q -value: $Q = (\sum m_{\text{react}} - \sum m_{\text{prod}}) \times 931.5 \text{ MeV}$

Fission & Fusion

- Fission: $^{235}\text{U} \rightarrow \text{fragments} + \text{neutrons} + \sim 200 \text{ MeV}$
- Fusion: $4\text{ }^1\text{H} \rightarrow\text{}^4\text{He} + 2e^+ + 2\nu + 26.7 \text{ MeV}$
- Fusion requires $\sim 10^7 \text{ K}$ (thermonuclear)

Tip: For definite integrals, check if the function is even or odd — this can halve your work.

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19. Semiconductor Electronics

Band Theory & Semiconductors

- Conductors: no gap; Semiconductors: $E_g = 0.1\text{--}3\text{ eV}$; Insulators: $E_g > 3\text{ eV}$
- Si: $E_g = 1.1\text{ eV}$; Ge: $E_g = 0.67\text{ eV}$
- n-type: pentavalent (P, As); majority = e^-
- p-type: trivalent (B, Al, Ga); majority = h^+
- Mass action: $n_e \cdot n_h = n_i^2$

p-n Junction & Diode

- Barrier: Si $\approx 0.7\text{ V}$; Ge $\approx 0.3\text{ V}$
- Forward bias: narrow depletion, large current
- Reverse bias: wide depletion, tiny current (I_0)
- $I = I_0(e^{eV/k_B T} - 1)$
- Half-wave rectifier: $f_{\text{out}} = f_{\text{in}}$
- Full-wave rectifier: $f_{\text{out}} = 2f_{\text{in}}$

Special Diodes

- Zener: reverse breakdown, voltage regulation ($V_{\text{out}} = V_Z$)
- LED: forward bias, $E_{\text{photon}} \approx E_g$
- Photodiode: reverse bias, current $\propto$ light intensity
- Solar cell: no external bias, photovoltaic effect

Transistor

- $I_E = I_B + I_C$; $\alpha = I_C/I_E$; $\beta = I_C/I_B$
- $\beta = \alpha/(1 - \alpha)$; $\alpha = \beta/(1 + \beta)$
- Active mode: EB forward, CB reverse
- CE voltage gain: $A_V = -\beta R_C/R_B$; phase inversion
- Switch: cutoff (OFF), saturation (ON)

Logic Gates

Gate	Expression	Output 1 when
OR	$A + B$	any input 1
AND	$A \cdot B$	all inputs 1
NOT	$\bar{A}$	input 0
NAND	$\overline{A \cdot B}$	not all 1
NOR	$\overline{A + B}$	all inputs 0
XOR	$A \oplus B$	inputs differ

De Morgan: $\overline{A \cdot B} = \bar{A} + \bar{B}$; $\overline{A + B} = \bar{A} \cdot \bar{B}$

NAND and NOR are universal gates.

PART II

Chemistry

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20. Mole Concept

Fundamental Relations

- 1 mole = 6.022×10^{23} particles (N_A)
- Moles: $n = \text{mass}/M = N/N_A$
- Gas at STP (0°C , 1 atm): 1 mol = 22.4 L
- Molarity (M) = $n_{\text{solute}}/V_{\text{soln}}$ (L)
- Molality (m) = $n_{\text{solute}}/W_{\text{solvent}}$ (kg)
- Normality: $N = n \times n\text{-factor}/V_L$; $N_1 V_1 = N_2 V_2$

Stoichiometry

- Balance equation; use mole ratio
- Limiting reagent: one consumed first
- % yield = (actual/theoretical) $\times 100$
- % composition = (mass of element/ M) $\times 100$
- Empirical formula $\rightarrow$ Molecular: $n = M_{\text{mol}}/M_{\text{emp}}$

Equivalent Weight

- Acid: $M/\text{basicity}$; Base: $M/\text{acidity}$
- Redox: $M/\text{change in ON per formula unit}$
- $n\text{-factor}$ = no. of $H^+/\text{OH}^-/\text{electrons transferred}$

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21. Atomic Structure

Key Formulas & Numbers

- Bohr: $E_n = -13.6Z^2/n^2 \text{ eV}$; $r_n = 0.529n^2/Z \text{ \AA}$
- de Broglie: $\lambda = h/(mv)$; $h = 6.626 \times 10^{-34} \text{ J s}$
- Heisenberg: $\Delta x \cdot \Delta p \geq h/(4\pi)$
- Schrödinger: $\hat{H}\psi = E\psi$ (wave equation)

Quantum Numbers

- $n = 1, 2, 3, \dots$ (shell); $l = 0, 1, \dots, n-1$ (subshell: s,p,d,f)
- $m_l = -l$ to $+l$ (orbitals); $m_s = \pm \frac{1}{2}$ (spin)
- Max e^- in shell: $2n^2$; in subshell: $2(2l + 1)$
- s:2, p:6, d:10, f:14
- Aufbau: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d...
- Exceptions: Cr = $[\text{Ar}]3d^5 4s^1$; Cu = $[\text{Ar}]3d^{10} 4s^1$

Rules

- Aufbau: fill lowest energy first ($n + l$ rule)
- Hund: max unpaired electrons in degenerate orbitals
- Pauli: no two electrons with same 4 quantum numbers

Tip: $|AB| = |A| \cdot |B|$ and $(AB)^{-1} = B^{-1}A^{-1}$ solve most JEE matrix problems.



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22. Chemical Bonding

VSEPR & Hybridization

SN	Hybrid	Geometry	Angle
2	sp	Linear	180°
3	sp^2	Trig. planar	120°
4	sp^3	Tetrahedral	109.5°
5	sp^3d	TBP	$90^\circ, 120^\circ$
6	sp^3d^2	Octahedral	90°

SN = steric number = BP + LP

Common Shapes with Lone Pairs

BP	LP	Shape	Example
3	1	Pyramidal	NH ₃
2	2	Bent/V	H ₂ O
4	1	See-saw	SF ₄
4	2	Square planar	XeF ₄
3	2	T-shape	ClF ₃
2	3	Linear	XeF ₂
5	1	Square pyramidal	BrF ₅

Bond Properties

- Bond order = (bonding e^- – antibonding e^-)/2
- Higher BO $\rightarrow$ shorter, stronger bond
- Formal charge = $V - L - B/2$
- Dipole moment $\mu = q \times d$
- Fajan's rules: small cation + large anion $\rightarrow$ covalent character

MO Theory (Diatomic)

- O₂: paramagnetic (2 unpaired), BO = 2
- N₂: diamagnetic, BO = 3 (triple bond)
- For $Z \leq 7$: σ_{2p} above π_{2p} ; For $Z \geq 8$: σ_{2p} below π_{2p}

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23. States of Matter

Gas Laws

- Ideal: $PV = nRT$; $R = 8.314 \text{ J/mol K} = 0.0821 \text{ L atm/mol K}$
- Boyle: $P_1 V_1 = P_2 V_2$ (const. T)
- Charles: $V_1/T_1 = V_2/T_2$ (const. P)
- Gay-Lussac: $P_1/T_1 = P_2/T_2$ (const. V)
- Dalton: $P_{\text{total}} = \sum p_i$; $p_i = x_i P$
- Graham: $r_1/r_2 = \sqrt{M_2/M_1}$
- $v_{\text{rms}} = \sqrt{3RT/M}$; $KE = \frac{3}{2} nRT$

Real Gases

- van der Waals: $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$
- a : intermolecular attraction; b : molecular volume
- Boyle temperature: $T_B = a/(Rb)$
- Critical: $T_c = 8a/(27Rb)$; $P_c = a/(27b^2)$; $V_c = 3nb$
- Compressibility factor: $Z = PV/(nRT)$; ideal: $Z = 1$



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24. Chemical Thermodynamics

Key Formulas

- First law: $\Delta U = q + w$; $w = -P_{\text{ext}}\Delta V$ (expansion)
- At const. P : $q_P = \Delta H$; At const. V : $q_V = \Delta U$
- $\Delta H = \Delta U + \Delta n_g RT$
- Hess: $\Delta H_{\text{rxn}} = \sum \Delta H_f^\circ(\text{prod}) - \sum \Delta H_f^\circ(\text{react})$
- Bond energy: $\Delta H = \sum BE(\text{broken}) - \sum BE(\text{formed})$
- Kirchhoff: $\Delta H(T_2) = \Delta H(T_1) + \Delta C_p(T_2 - T_1)$

Entropy & Free Energy

- $\Delta S = q_{\text{rev}}/T$; $\Delta S_{\text{universe}} \geq 0$
- Gibbs: $\Delta G = \Delta H - T\Delta S$
- Spontaneous: $\Delta G < 0$; Equilibrium: $\Delta G = 0$
- $\Delta G^\circ = -RT \ln K = -nFE^\circ$
- $\Delta G = \Delta G^\circ + RT \ln Q$

Tip: For complex numbers, use polar form for multiplication and rectangular for addition.

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25. Equilibrium

Chemical Equilibrium

- $K_c = [\text{products}]^p / [\text{reactants}]^r$ (at equilibrium)
- $K_p = K_c (RT)^{\Delta n_g}$
- Reaction quotient Q : if $Q < K$ forward; $Q > K$ backward; $Q = K$ equilibrium
- Le Chatelier: system shifts to oppose the change
- $\Delta G^\circ = -RT \ln K$

Ionic Equilibrium

- $K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$ (25°C)
- $\text{pH} = -\log[\text{H}^+]$; $\text{pOH} = -\log[\text{OH}^-]$; $\text{pH} + \text{pOH} = 14$
- Weak acid: $[\text{H}^+] = \sqrt{K_a C}$; $\alpha = \sqrt{K_a / C}$
- Weak base: $[\text{OH}^-] = \sqrt{K_b C}$; $K_a K_b = K_w$
- Henderson: $\text{pH} = \text{p}K_a + \log([\text{salt}]/[\text{acid}])$
- Solubility product: K_{sp} ; precipitation if $\text{IP} > K_{sp}$
- Common ion effect reduces solubility/ionization



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26. Redox & Electrochemistry

Key Formulas

- Oxidation: loss of e^- , increase in ON
- Reduction: gain of e^- , decrease in ON
- $E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$
- Nernst: $E = E^\circ - \frac{0.0591}{n} \log Q$ (25°C)
- $\Delta G^\circ = -nFE^\circ$; $F = 96485 \text{ C/mol}$
- At equilibrium: $E = 0$; $\log K = nE^\circ / 0.0591$

Electrolysis

- Faraday I: $m = ZIt$; $Z = M/(nF)$
- Faraday II: $m_1/m_2 = E_1/E_2$ (same charge)
- Conductivity: $\kappa = 1/\rho = G \times l/A$
- Molar conductivity: $\Lambda_m = \kappa \times 1000/c$ ($\text{S cm}^2/\text{mol}$)
- Kohlrausch: $\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$

- Λ_m increases on dilution; strong electrolyte: slight; weak: large

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27. Chemical Kinetics

Rate Laws & Integrated

- Rate = $k[A]^m[B]^n$; Order = $m + n$
- Zero: $[A] = [A]_0 - kt$; $t_{1/2} = [A]_0 / (2k)$
- First: $k = \frac{1}{t} \ln \frac{[A]_0}{[A]} = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$; $t_{1/2} = 0.693/k$
- Second: $1/[A] = 1/[A]_0 + kt$; $t_{1/2} = 1/(k[A]_0)$
- $t_{1/2} \propto [A]_0^{1-n}$ for n th order

Arrhenius & Activation

- $k = Ae^{-E_a/(RT)}$; $\ln k = \ln A - E_a/(RT)$
- $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$
- Catalyst lowers E_a ; increases rate without being consumed
- $E_a(\text{forward}) + \Delta H = E_a(\text{backward})$ (exothermic: $\Delta H < 0$)

Tip: SN1: tertiary + polar protic. SN2: primary + polar aprotic. This is tested frequently.



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28. Solutions & Colligative Properties

Key Formulas

- Raoult: $P = x_{\text{solvent}}P^\circ$; $\Delta P/P^\circ = x_{\text{solute}}$
- Positive deviation: $P > \text{Raoult}$ (weaker interactions)
- Negative deviation: $P < \text{Raoult}$ (stronger interactions)
- Henry: $p = K_H x$ (dissolved gas)
- Colligative properties depend on number of solute particles:

Colligative Properties

- Relative lowering of VP: $\Delta P/P^\circ = i \cdot x_2$
- Boiling point elevation: $\Delta T_b = iK_b m$
- Freezing point depression: $\Delta T_f = iK_f m$
- Osmotic pressure: $\pi = iCRT = i(n/V)RT$
- van't Hoff factor: $i = 1 + (n - 1)\alpha$ (electrolytes)
- $i > 1$: dissociation; $i < 1$: association

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29. Surface Chemistry**Adsorption & Catalysis**

- Physisorption: van der Waals, reversible, multilayer, low E_a
- Chemisorption: chemical bonds, irreversible, monolayer, high E_a
- Freundlich: $x/m = kP^{1/n}$; Langmuir: monolayer saturation model
- Catalyst: lowers E_a ; homogeneous (same phase) / heterogeneous

Colloids

- Particle size: 1–1000 nm
- Tyndall effect, Brownian motion, electrophoresis
- Lyophilic (reversible): starch, gum; Lyophobic (irreversible): metal sols
- Coagulation: Hardy-Schulze (higher valence ion → better coag.)
- Emulsions: o/w (milk) or w/o (butter); emulsifier needed
- Protective colloids: lyophilic protects lyophobic (gold number)



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30. Periodic Table & Periodicity

Periodic Trends

Property	Across period	Down group
Atomic radius	Decreases	Increases
IE	Increases	Decreases
EA	Generally increases	Generally decreases
EN	Increases	Decreases
Metallic character	Decreases	Increases

Key Points

- IE anomalies: $\text{Be} > \text{B}$ (2s vs 2p); $\text{N} > \text{O}$ (half-filled stability)
- EA: $\text{Cl} > \text{F}$ (small size of F, electron repulsion)
- EN scale: Pauling ($F = 4.0$, most EN)
- Diagonal relationship: $\text{Li} \sim \text{Mg}$, $\text{Be} \sim \text{Al}$, $\text{B} \sim \text{Si}$
- Oxidation states: max = group number (for main group)

Tip: Order of reaction is determined experimentally — never from the stoichiometric equation.

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31. s-Block Elements

Key Facts

- Group 1: ns^1 ; +1 OS; strong reducing agents; reactivity $\uparrow$ down
- Group 2: ns^2 ; +2 OS; Be is amphoteric (diagonal with Al)
- $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$
- NaOH: caustic soda; Na_2CO_3 : washing soda; NaHCO_3 : baking soda
- CaO: quicklime; $\text{Ca}(\text{OH})_2$: slaked lime
- $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$: gypsum; $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$: Plaster of Paris
- Li: anomalous (small size, high IE); diagonal with Mg
- Flame colors: Li-red, Na-yellow, K-violet, Ca-brick red, Sr-crimson, Ba-green



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32. p-Block Elements

Group 13 (Boron family)

- B: electron deficient; B_2H_6 (3c-2e bonds); BF_3 (sp^2 , Lewis acid)
- Al_2O_3 : amphoteric; Alums: $M^+M^{3+}(SO_4)_2 \cdot 12H_2O$

Group 14 (Carbon family)

- C allotropes: diamond (sp^3), graphite (sp^2), fullerene (C_{60})
- CO: neutral oxide, reducing agent; CO_2 : acidic oxide
- Si: semiconductor; SiO_2 (3D network, high m.p.)

Group 15 (Nitrogen family)

- N_2 : triple bond, inert; NH_3 : pyramidal, basic, sp^3
- HNO_3 : strong oxidizing acid
- P: white (P_4), red (polymer); PCl_3 (sp^3), PCl_5 (sp^3d)
- H_3PO_4 : tribasic; H_3PO_3 : dibasic; H_3PO_2 : monobasic

Group 16 (Oxygen family)

- O_3 : powerful oxidizer, bent
- H_2SO_4 : strong diprotic acid, dehydrating, oxidizing
- Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (V_2O_5 catalyst)
- SO_2 : acidic, reducing agent, bleaching (temporary)

Group 17 (Halogens)

- Reactivity: $F_2 > Cl_2 > Br_2 > I_2$
- HF: weak acid (H-bonding); HCl, HBr, HI: strong
- Bleaching powder: $Ca(OCl)Cl$; Interhalogen: ClF_3 , BrF_5 , IF_7

Group 18 (Noble gases)

- XeF_2 : linear (sp^3d); XeF_4 : square planar (sp^3d^2)
- XeF_6 : distorted octahedral; XeO_3 : pyramidal

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33. d- and f-Block Elements

d-Block Properties

- Config: $(n-1)d^{1-10}ns^{0-2}$; variable OS, colored ions, catalytic
- Cr: $[Ar]3d^5 4s^1$; Cu: $[Ar]3d^{10} 4s^1$ (stability of half/full d)
- $KMnO_4$: strong oxidizer ($Mn^{7+} \rightarrow Mn^{2+}$ in acid)
- $K_2Cr_2O_7$: oxidizer in acidic medium ($Cr^{6+} \rightarrow Cr^{3+}$)
- Magnetic moment: $\mu = \sqrt{n(n+2)}$ BM (n = unpaired e^-)

f-Block

- Lanthanides (4f): +3 common OS; lanthanide contraction
- Actinides (5f): variable OS; many radioactive

Tip: Identify the functional group first, then the reagent. The group determines the reaction type.



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**34. Coordination Compounds****Key Concepts**

- CN = no. of donor atoms bonded to metal
- IUPAC: ligands (alphabetical, anionic suffix -o) then metal (-ate for anion)
- Isomers: geometrical (cis/trans), optical, linkage, ionization, hydrate
- CFT: octahedral splits into t_{2g} (lower) and e_g (upper); Δ_o
- Tetrahedral: $\Delta_t = (4/9)\Delta_o$ (always high spin)
- Strong field (large Δ): low spin; Weak field: high spin
- Spectrochemical: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$
- CFSE = $(-0.4\Delta_o \times n_{t_{2g}}) + (0.6\Delta_o \times n_{e_g})$

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35. Metallurgy

Key Processes

- Concentration: gravity, magnetic, froth flotation, leaching
- Reduction: C (Fe, Zn, Sn), Al/thermite (Cr, Mn), electrolysis (Al, Na, Mg)
- Ellingham: lower ΔG° line = more stable oxide
- Fe: blast furnace; $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
- Al: Hall-Heroult; electrolysis of Al_2O_3 in cryolite (Na_3AlF_6)
- Cu: matte smelting; $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$; electrolytic refining
- Refining: electrolytic, zone (Si, Ge), van Arkel (Ti, Zr), liquation (Sn)



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36. General Organic Chemistry

Effects & Stability

- $-I$: F, Cl, Br, NO_2 , CN, COOH, COR; $+I$: alkyl groups ($3^\circ > 2^\circ > 1^\circ$)
- $+M$: OH, OR, NH_2 , NR_2 ; $-M$: NO_2 , CHO, COR, COOH, CN
- Hyperconjugation: no-bond resonance; stabilizes carbocations & alkenes
- Carbocation stability: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$; allyl $\approx 2^\circ$
- Carbanion stability: opposite; stabilized by $-I/-M$ groups
- Free radical: $3^\circ > 2^\circ > 1^\circ$

Acidity & Basicity

- Acidity increases with $-I$ groups, resonance stabilization of conjugate base
- Acid strength: $\text{RCOOH} > \text{ArOH} > \text{ROH} > \text{H}_2\text{O} > \text{alkyne} > \text{NH}_3 > \text{alkane}$
- Basicity: amines $> \text{NH}_3 > \text{arylamines} > \text{amides}$

Reaction Types

- Electrophilic addition: alkenes, alkynes
- Electrophilic aromatic substitution: benzene
- Nucleophilic substitution: SN_1 , SN_2 (haloalkanes)
- Nucleophilic addition: carbonyl compounds
- Elimination: E_1 , E_2 (dehydrohalogenation)
- Free radical: alkane halogenation
- Aromaticity: planar, cyclic, $(4n + 2)\pi$ (Hückel)

Tip: If $E_{\text{cell}}^{\circ} > 0$, the reaction is spontaneous. Connect to $\Delta G^{\circ} = -nFE^{\circ}$.

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37. Hydrocarbons

Alkanes

- C_nH_{2n+2} ; sp^3 ; free radical substitution (halogenation)
- Reactivity: $F_2 > Cl_2 > Br_2$; $3^{\circ}H > 2^{\circ}H > 1^{\circ}H$

Alkenes

- C_nH_{2n} ; sp^2 ; electrophilic addition
- Markownikoff: H adds to C with more H's
- Anti-Markownikoff: HBr + peroxide (radical mechanism)
- Ozonolysis: $C=C \xrightarrow{O_3/Zn}$ aldehydes/ketones
- $KMnO_4$ (cold, dilute): syn-dihydroxylation (Baeyer's test)

Alkynes & Arenes

- C_nH_{2n-2} ; sp ; terminal alkynes are acidic ($pK_a \approx 25$)
- Na/liq. NH_3 : trans-alkene (Birch-type); H_2 /Lindlar: cis-alkene
- Benzene: electrophilic aromatic substitution (EAS)
- EAS: nitration (NO_2^+), halogenation (X_2/FeX_3), Friedel-Crafts ($RX/AlCl_3$)
- Activating (o/p): OH, NH_2 , OR, alkyl; Deactivating (m): NO_2 , COOH, CHO, CN
- Halogens: o/p directing but deactivating



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38. Haloalkanes & Haloarenes

Substitution & Elimination

- SN_2 : one step, backside attack, Walden inversion; rate = $k[\text{sub}][\text{Nu}]$; 1° best
- SN_1 : two steps, carbocation, racemization; rate = $k[\text{sub}]$; 3° best
- E_2 : strong base, anti-periplanar, Saytzeff (more substituted alkene)
- E_1 : weak base, carbocation intermediate; 3° best

- Competition: strong base $\rightarrow$ E2; weak base/nucleophile $\rightarrow$ SN1 (3°) or SN2 (1°)

Named Reactions

- Wurtz: $2RX + 2Na \rightarrow R-R + 2NaX$
- Finkelstein: $RCl + NaI \text{ (acetone)} \rightarrow RI + NaCl$
- Swarts: $RCl + AgF \rightarrow RF + AgCl$
- Haloarenes: nucleophilic substitution difficult (C-X bond has partial double bond character)

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39. Alcohols, Phenols & Ethers

Key Reactions

- Acidity: $ArOH > H_2O > ROH$
- Lucas test: 3° (instant), 2° (5 min), 1° (no reaction at RT)
- Oxidation: $1^\circ ROH \xrightarrow{PCC} RCHO \xrightarrow{[O]} RCOOH$; $2^\circ \xrightarrow{[O]} \text{ketone}$
- Dehydration: $ROH \xrightarrow{H_2SO_4, \Delta} \text{alkene (Saytzeff)}$
- Williamson: $RONa + R'X \rightarrow ROR' + NaX$ (1° RX preferred)
- Phenol: Kolbe (salicylic acid), Reimer-Tiemann (salicylaldehyde)
- Phenol: $FeCl_3$ test (violet color)

Tip: Remember exceptions: Cr is $[Ar]3d^5 4s^1$ and Cu is $[Ar]3d^{10} 4s^1$.



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40. Aldehydes, Ketones & Acids

Carbonyl Reactions

- Nucleophilic addition at $C=O$
- Aldol: $2RCHO \xrightarrow{NaOH} \beta\text{-hydroxy aldehyde} \xrightarrow{\Delta} \alpha, \beta\text{-unsaturated}$
- Cross-Aldol: different aldehydes with NaOH
- Cannizzaro: $HCHO \text{ (no } \alpha\text{-H)} \xrightarrow{\text{conc. NaOH}} HCOO^- + CH_3OH$
- Tollens: Ag mirror test ($RCHO$ only)

- Fehling: Cu_2O red ppt (RCHO , not ArCHO)
- Iodoform: $\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CHOH}- \xrightarrow{\text{I}_2/\text{NaOH}} \text{CHI}_3$

Reduction & Named Reactions

- Clemmensen: $\text{Zn-Hg/HCl} \rightarrow \text{CH}_2$ (acidic)
- Wolff-Kishner: $\text{NH}_2\text{NH}_2/\text{KOH} \rightarrow \text{CH}_2$ (basic)
- Wittig: $\text{Ph}_3\text{P=CHR} + \text{RCOR} \rightarrow \text{alkene}$
- HVZ: α -bromination of RCOOH (Br_2/P)
- Decarboxylation: $\text{RCOONa} \xrightarrow{\text{NaOH/CaO}} \text{RH}$
- Ester: $\text{RCOOH} + \text{R'OH} \rightleftharpoons \text{RCOOR'} + \text{H}_2\text{O}$ (Fischer)

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41. Amines

Key Points

- Basicity (gas): $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$
- Basicity (aq): $2^\circ > 1^\circ > 3^\circ > \text{NH}_3$ (solvation effect)
- $\text{ArNH}_2 < \text{NH}_3$ (resonance delocalizes lone pair)
- Carbylamine: 1° amine + $\text{CHCl}_3/\text{KOH} \rightarrow$ isocyanide (bad smell)
- Hinsberg: ArSO_2Cl distinguishes 1° , 2° , 3°
- Diazotization: $\text{ArNH}_2 + \text{NaNO}_2/\text{HCl}$ ($0-5^\circ\text{C}$) $\rightarrow \text{ArN}_2^+\text{Cl}^-$
- Sandmeyer: $\text{ArN}_2^+ + \text{CuX} \rightarrow \text{ArX}$ (Cl, Br, CN)
- Coupling: $\text{ArN}_2^+ + \text{phenol/amine} \rightarrow \text{azo dye}$
- Gabriel synthesis: pure 1° amines (phthalimide route)
- Hofmann: amide $\xrightarrow{\text{Br}_2/\text{NaOH}} 1^\circ$ amine (one fewer C)



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42. Biomolecules & Polymers

Carbohydrates

- Glucose: $\text{C}_6\text{H}_{12}\text{O}_6$; α -D and β -D forms
- Sucrose: glucose + fructose; non-reducing (no free anomeric C)
- Maltose (glucose+glucose), Lactose (galactose+glucose): reducing

- Starch: amylose (α -1,4) + amylopectin (α -1,4 and α -1,6)
- Cellulose: β -1,4 glucose polymer

Amino Acids & Proteins

- α -amino acids; zwitter ion at isoelectric pH
- Peptide bond: $-\text{CO}-\text{NH}-$ (amide linkage)
- 1°: sequence; 2°: α -helix/ β -sheet; 3°: folding; 4°: subunits
- Denaturation: loss of 2°/3°/4° structure

Nucleic Acids

- DNA: A-T (2 H-bonds), G-C (3 H-bonds); deoxyribose sugar; double helix
- RNA: single strand; ribose sugar; U replaces T

Polymers

- Addition: polyethylene, PVC ($\text{CHCl}=\text{CH}_2$), Teflon ($\text{CF}_2=\text{CF}_2$), polystyrene
- Condensation: nylon-6,6, nylon-6, polyester (terylene), bakelite (phenol+HCHO)
- Natural rubber: cis-polyisoprene; vulcanization with S (cross-linking)
- Biodegradable: PHBV, nylon-2-nylon-6
- Elastomers (rubber), fibres (nylon), thermoplastics (PE), thermosets (bakelite)

Tip: In mole concept problems, always start by writing the balanced equation.

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43. Chemistry in Everyday Life

Drugs & Chemicals

- Analgesics: aspirin (also antipyretic), paracetamol, ibuprofen
- Antibiotics: penicillin, amoxicillin; bactericidal vs bacteriostatic
- Antiseptics: dettol, boric acid (on living tissue)
- Disinfectants: Cl_2 , SO_2 , phenol 1% (non-living surfaces)
- Antacids: NaHCO_3 , $\text{Mg}(\text{OH})_2$, $\text{Al}(\text{OH})_3$
- Antihistamines: brompheniramine
- Tranquilizers: valium, chlordiazepoxide

Other

- Soaps: RCOONa ; fail in hard water ($\text{Ca}^{2+}/\text{Mg}^{2+}$ ppt)
- Detergents: RSO_3Na ; work in hard water

- Artificial sweeteners: saccharin, aspartame, sucralose
- Food preservatives: Na-benzoate, KMS (potassium metabisulfite)
- Biodegradable vs non-biodegradable detergents

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44. Sets, Relations & Functions

Sets

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B \cup C) = \sum n - \sum n(\text{pairs}) + n(A \cap B \cap C)$
- De Morgan: $(A \cup B)' = A' \cap B'$; $(A \cap B)' = A' \cup B'$
- Power set: $|P(A)| = 2^{|A|}$
- $A - B = A \cap B'$

Relations & Functions

- Reflexive: $(a, a) \in R \forall a$; Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$
- Transitive: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$
- Equivalence: reflexive + symmetric + transitive
- Injective (1-1): $f(a) = f(b) \Rightarrow a = b$
- Surjective (onto): range = codomain
- Bijective: 1-1 and onto; inverse exists
- Even: $f(-x) = f(x)$; Odd: $f(-x) = -f(x)$
- $(f \circ g)(x) = f(g(x))$

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45. Complex Numbers

Key Formulas

- $z = a + ib$; $|z| = \sqrt{a^2 + b^2}$; $\arg z = \tan^{-1}(b/a)$
- $\bar{z} = a - ib$; $z\bar{z} = |z|^2$; $z + \bar{z} = 2\text{Re}(z)$
- Polar: $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$
- De Moivre: $(re^{i\theta})^n = r^n e^{in\theta}$
- n th roots of unity: $e^{2\pi i k/n}$, $k = 0, \dots, n-1$; sum = 0
- Cube roots: $1, \omega, \omega^2$; $\omega = e^{2\pi i/3}$; $1 + \omega + \omega^2 = 0$; $\omega^3 = 1$
- Triangle inequality: $||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$
- $|z_1 z_2| = |z_1| |z_2|$; $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
- Rotation: $z' = z_0 + (z - z_0)e^{i\theta}$

Tip: In optics, follow the sign convention consistently. Most errors come from sign mistakes.



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46. Quadratic Equations

Key Formulas

- $ax^2 + bx + c = 0$: $x = \frac{-b \pm \sqrt{D}}{2a}$; $D = b^2 - 4ac$
- $D > 0$: real distinct; $D = 0$: equal; $D < 0$: complex conjugate
- Sum: $\alpha + \beta = -b/a$; Product: $\alpha\beta = c/a$
- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- $|\alpha - \beta| = \sqrt{D}/|a|$
- For $ax^2 + bx + c > 0 \forall x$: $a > 0$ and $D < 0$
- If D is perfect square and $a, b, c \in \mathbb{Q}$: roots are rational
- Cubic: $\alpha + \beta + \gamma = -b/a$; $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$; $\alpha\beta\gamma = -d/a$

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47. Sequences & Series

Arithmetic Progression

- $a_n = a + (n - 1)d$; $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$
- n th term from S_n : $a_n = S_n - S_{n-1}$
- AM of a, b : $(a + b)/2$
- Insert n AMs between a, b : $d = (b - a)/(n + 1)$

Geometric Progression

- $a_n = ar^{n-1}$; $S_n = a(r^n - 1)/(r - 1)$
- $S_\infty = a/(1 - r)$ for $|r| < 1$
- GM of a, b : $\sqrt{ab}$; Insert n GMs: $r = (b/a)^{1/(n+1)}$

Special Series & Inequalities

- $\sum_{k=1}^n k = n(n+1)/2$
- $\sum k^2 = n(n+1)(2n+1)/6$
- $\sum k^3 = [n(n+1)/2]^2$
- $AM \geq GM \geq HM$ (for positive numbers; equality iff all equal)
- HM of a, b : $2ab/(a+b)$
- AGP: $S = a/(1-r) + dr/(1-r)^2$ for $|r| < 1$



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48. Binomial Theorem

Key Formulas

- $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$
- $T_{r+1} = \binom{n}{r} a^{n-r} b^r$
- Middle term: $T_{n/2+1}$ (n even); two middle (n odd)
- Greatest term: find r where $T_{r+1}/T_r \geq 1$ switches
- $\binom{n}{r} = \binom{n}{n-r}$; $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$
- $\sum_{r=0}^n \binom{n}{r} = 2^n$; $\sum (-1)^r \binom{n}{r} = 0$
- $(1+x)^n \approx 1 + nx$ for $|x| \ll 1$ (any n)
- General: $(1+x)^n = \sum_{r=0}^{\infty} \binom{n}{r} x^r$ for $|x| < 1$, any real n

Tip: For AC circuits: current leads voltage in capacitors, voltage leads current in inductors (CIVIL).

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49. Permutations & Combinations

Key Formulas

- $P(n, r) = n!/(n-r)!;$ $C(n, r) = n!/[r!(n-r)!]$
- Circular: $(n-1)!;$ Necklace: $(n-1)!/2$
- With repetition: arrangements $= n^r$; selections $= \binom{n+r-1}{r}$
- Identical objects: $n!/(n_1!n_2!\dots)$
- Derangements: $D_n = n! \sum_{k=0}^n (-1)^k/k!$

- Division into groups: $\frac{n!}{(r!)^k} \cdot \frac{1}{k!}$ (if groups identical)
- $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$



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50. Matrices & Determinants

Matrix Properties

- $(AB)^T = B^T A^T$; $(AB)^{-1} = B^{-1} A^{-1}$
- $|AB| = |A||B|$; $|kA| = k^n |A|$ ($n \times n$)
- $|A^{-1}| = 1/|A|$; $|A^T| = |A|$
- $A(\text{adj } A) = |A|I$; $A^{-1} = \text{adj } A / |A|$
- $|\text{adj } A| = |A|^{n-1}$

Determinants

- 2×2 : $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
- 3×3 : cofactor expansion along any row/column
- Row/column operations don't change $|A|$ (for evaluation)
- $R_i \leftrightarrow R_j$: sign changes; $R_i \rightarrow kR_i$: multiply by k
- Cramer: $x = D_x/D$, $y = D_y/D$, $z = D_z/D$
- System consistent if $D \neq 0$ (unique solution)

Types

- Symmetric: $A^T = A$; Skew-symmetric: $A^T = -A$; Orthogonal: $AA^T = I$
- Idempotent: $A^2 = A$; Involutory: $A^2 = I$; Nilpotent: $A^k = O$

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51. Limits, Continuity & Differentiability

Standard Limits

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$; $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$; $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1$

- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$; $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$; $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$
- $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$; $\lim_{x \rightarrow \infty} (1+1/x)^x = e$
- $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$
- L'Hôpital: $0/0$ or ∞/∞ : $\lim f/g = \lim f'/g'$

Continuity & Differentiability

- Continuous at a : $\lim_{x \rightarrow a^-} f = \lim_{x \rightarrow a^+} f = f(a)$
- Differentiable $\Rightarrow$ continuous (converse false)
- $|x|$: continuous everywhere, not differentiable at $x = 0$
- $[x]$ (floor): discontinuous at integers

Tip: In thermodynamics, always identify the process type before applying formulas.



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52. Differentiation & Applications

Standard Derivatives

- $\frac{d}{dx} x^n = nx^{n-1}$; $\frac{d}{dx} e^x = e^x$; $\frac{d}{dx} a^x = a^x \ln a$
- $\frac{d}{dx} \ln x = 1/x$; $\frac{d}{dx} \log_a x = 1/(x \ln a)$
- $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$; $(\tan x)' = \sec^2 x$
- $(\sec x)' = \sec x \tan x$; $(\csc x)' = -\csc x \cot x$
- $(\sin^{-1} x)' = 1/\sqrt{1-x^2}$; $(\tan^{-1} x)' = 1/(1+x^2)$

Rules

- Chain: $(f(g(x)))' = f'(g(x)) \cdot g'(x)$
- Product: $(uv)' = u'v + uv'$; Quotient: $(u/v)' = (u'v - uv')/v^2$
- Logarithmic: $y = f^g \Rightarrow \ln y = g \ln f$, then differentiate
- Parametric: $dy/dx = (dy/dt)/(dx/dt)$
- Implicit: differentiate both sides w.r.t. x

Applications

- Tangent slope at (x_0, y_0) : $m = f'(x_0)$; Normal slope: $-1/m$
- Increasing: $f' > 0$; Decreasing: $f' < 0$
- Local max: $f' = 0, f'' < 0$; Local min: $f' = 0, f'' > 0$
- Inflection: $f'' = 0$ and changes sign
- Rolle: $f(a) = f(b)$, continuous, differentiable $\Rightarrow \exists c : f'(c) = 0$
- LMVT: $\exists c : f'(c) = [f(b) - f(a)]/(b - a)$
- Rate of change: $dy/dt = (dy/dx)(dx/dt)$

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53. Integrals**Standard Integrals**

- $\int x^n dx = x^{n+1}/(n+1) + C$; $\int dx/x = \ln|x| + C$
- $\int e^x dx = e^x$; $\int a^x dx = a^x/\ln a$
- $\int \sin x dx = -\cos x$; $\int \cos x dx = \sin x$
- $\int \sec^2 x dx = \tan x$; $\int \csc^2 x dx = -\cot x$
- $\int \sec x \tan x dx = \sec x$; $\int \csc x \cot x dx = -\csc x$
- $\int dx/\sqrt{1-x^2} = \sin^{-1} x$; $\int dx/(1+x^2) = \tan^{-1} x$
- $\int dx/\sqrt{x^2+a^2} = \ln|x + \sqrt{x^2+a^2}|$
- $\int dx/(x^2-a^2) = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$

Techniques

- By parts: $\int uv' dx = uv - \int u'v dx$; ILATE priority
- $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$
- Partial fractions for rational functions
- $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}(x/a)$

Definite Integral Properties

- King: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
- Even: $\int_{-a}^a f = 2 \int_0^a f$; Odd: $\int_{-a}^a f = 0$
- Periodic: $\int_0^{nT} f = n \int_0^T f$
- Walli's: $\int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx$
- Leibniz: $\frac{d}{dx} \int_{a(x)}^{b(x)} f(t) dt = f(b)b' - f(a)a'$



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54. Applications of Integrals

Area Formulas

- Under curve: $A = \int_a^b |f(x)| dx$
- Between curves: $A = \int_a^b |f(x) - g(x)| dx$
- Circle $x^2 + y^2 = r^2$: area = πr^2
- Ellipse $x^2/a^2 + y^2/b^2 = 1$: area = πab
- Parabola $y^2 = 4ax$, 0 to $x = a$: area = $\frac{4}{3}a^2$
- Between parabola $y^2 = 4ax$ and line $y = mx$: use intersection points

Tip: For problems involving springs and pulleys, draw free body diagrams first.

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55. Differential Equations

Types & Methods

- Order = highest derivative; Degree = power of highest order deriv.
- Variable separable: $\int f(y) dy = \int g(x) dx$
- Homogeneous: $dy/dx = F(y/x)$; put $y = vx$
- Linear: $dy/dx + Py = Q$; IF = $e^{\int P dx}$
- Solution: $y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$
- Exact: $Mdx + Ndy = 0$ if $\partial M/\partial y = \partial N/\partial x$
- Bernoulli: $dy/dx + Py = Qy^n$; substitute $v = y^{1-n}$

Standard Results

- Growth/decay: $dN/dt = kN \Rightarrow N = N_0 e^{kt}$
- Newton's cooling: $dT/dt = -k(T - T_s)$



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56. Coordinate Geometry

Straight Lines

- Distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Section: $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
- Centroid: $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$
- Slope: $m = (y_2 - y_1)/(x_2 - x_1) = \tan \theta$
- Point-slope: $y - y_1 = m(x - x_1)$
- General: $ax + by + c = 0$; $m = -a/b$
- Distance point to line: $|ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$
- Angle between lines: $\tan \alpha = |(m_1 - m_2)/(1 + m_1 m_2)|$
- Parallel: $m_1 = m_2$; Perpendicular: $m_1 m_2 = -1$
- Area of $\triangle$: $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Circles

- $(x - h)^2 + (y - k)^2 = r^2$
- General: $x^2 + y^2 + 2gx + 2fy + c = 0$; centre $(-g, -f)$; $r = \sqrt{g^2 + f^2 - c}$
- Tangent from (x_1, y_1) : length $= \sqrt{S_1}$ where $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$
- Tangent at (x_1, y_1) : $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
- Two circles: radical axis $S_1 - S_2 = 0$

Parabola

- $y^2 = 4ax$: vertex $(0, 0)$, focus $(a, 0)$, directrix $x = -a$
- Tangent: $ty = x + at^2$ (parametric, t = parameter)
- Normal: $y + tx = 2at + at^3$
- Focal chord: if one end at t , other at $-1/t$

Ellipse

- $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$); $e = \sqrt{1 - b^2/a^2}$; $b^2 = a^2(1 - e^2)$
- Foci: $(\pm ae, 0)$; Directrices: $x = \pm a/e$
- Sum of focal distances $= 2a$ (constant)
- Latus rectum $= 2b^2/a$

Hyperbola

- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$; $e = \sqrt{1 + b^2/a^2}$
- Foci: $(\pm ae, 0)$; Asymptotes: $y = \pm (b/a)x$
- | Diff of focal distances | $= 2a$
- Rectangular hyperbola: $xy = c^2$; $e = \sqrt{2}$

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57. 3D Geometry

Key Formulas

- Distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
- Section formula: analogous to 2D
- Direction ratios $a : b : c$; DCs: $l = a/\sqrt{\sum a^2}$ etc.; $l^2 + m^2 + n^2 = 1$
- Angle between lines: $\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$

Lines & Planes

- Line: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$; $\vec{r} = \vec{a} + \lambda \vec{b}$
- Plane: $ax + by + cz = d$; $\vec{r} \cdot \vec{n} = d$
- Point to plane: $|ax_1 + by_1 + cz_1 - d|/\sqrt{a^2 + b^2 + c^2}$
- Angle: line-plane $\sin \theta = |\vec{b} \cdot \vec{n}|/(|\vec{b}||\vec{n}|)$
- Plane-plane $\cos \theta = |\vec{n}_1 \cdot \vec{n}_2|/(|\vec{n}_1||\vec{n}_2|)$
- Skew lines dist: $d = |(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|/|\vec{b}_1 \times \vec{b}_2|$
- Coplanar lines: $[\vec{a}_2 - \vec{a}_1, \vec{b}_1, \vec{b}_2] = 0$

Tip: In projectile motion, horizontal and vertical components are independent. Analyze them separately.



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58. Vector Algebra

Key Formulas

- $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$; $\hat{a} = \vec{a}/|\vec{a}|$
- Dot: $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$
- Cross: $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$; $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
- $\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$; $\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a} \parallel \vec{b}$
- Triangle area = $\frac{1}{2} |\vec{a} \times \vec{b}|$; Parallelogram = $|\vec{a} \times \vec{b}|$
- Scalar triple: $[\vec{a}, \vec{b}, \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \text{volume of parallelepiped}$
- Coplanar iff $[\vec{a}, \vec{b}, \vec{c}] = 0$
- Projection of $\vec{a}$ on $\vec{b}$: $(\vec{a} \cdot \vec{b})/|\vec{b}|$

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

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59. Probability & Statistics

Probability

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A') = 1 - P(A)$
- Conditional: $P(A|B) = P(A \cap B)/P(B)$
- Independent: $P(A \cap B) = P(A) \cdot P(B)$
- Bayes: $P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$
- Total probability: $P(B) = \sum P(B|A_i)P(A_i)$

Random Variables

- Mean: $E(X) = \sum x_i P(x_i)$; $\text{Var}(X) = E(X^2) - [E(X)]^2$
- Binomial: $P(X = r) = \binom{n}{r} p^r q^{n-r}$; $\mu = np$; $\sigma^2 = npq$
- Bernoulli: single trial, $P(\text{success}) = p$

Statistics

- Mean $\bar{x} = \sum f_i x_i / \sum f_i$
- Variance $\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i} = \frac{\sum f_i x_i^2}{\sum f_i} - \bar{x}^2$
- SD = $\sqrt{\text{variance}}$; CV = $(\sigma/\bar{x}) \times 100$
- Combined: $\bar{x} = (n_1 \bar{x}_1 + n_2 \bar{x}_2) / (n_1 + n_2)$



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60. Trigonometry

Fundamental Identities

- $\sin^2 \theta + \cos^2 \theta = 1$; $1 + \tan^2 \theta = \sec^2 \theta$; $1 + \cot^2 \theta = \csc^2 \theta$
- $\sin(-\theta) = -\sin \theta$; $\cos(-\theta) = \cos \theta$

Addition & Double Angle

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
- $\sin 2A = 2 \sin A \cos A$; $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$
- $\tan 2A = 2 \tan A / (1 - \tan^2 A)$
- $\sin^2 A = (1 - \cos 2A)/2$; $\cos^2 A = (1 + \cos 2A)/2$

Product & Sum

- $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$
- $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
- $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$

Triangle & Solution

- Sine rule: $a / \sin A = b / \sin B = c / \sin C = 2R$
- Cosine rule: $c^2 = a^2 + b^2 - 2ab \cos C$
- Area = $\frac{1}{2} ab \sin C = \sqrt{s(s-a)(s-b)(s-c)}$; $s = (a+b+c)/2$
- $R = abc/(4\Delta)$; $r = \Delta/s$
- $r_1 = \Delta/(s-a)$; $r_2 = \Delta/(s-b)$; $r_3 = \Delta/(s-c)$

Inverse Trigonometric

- $\sin^{-1} x + \cos^{-1} x = \pi/2$; $\tan^{-1} x + \cot^{-1} x = \pi/2$
- $\sec^{-1} x + \csc^{-1} x = \pi/2$
- $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ ($xy < 1$)
- $2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2} = \cos^{-1} \frac{1-x^2}{1+x^2} = \tan^{-1} \frac{2x}{1-x^2}$
- Principal values: $\sin^{-1} \in [-\pi/2, \pi/2]$; $\cos^{-1} \in [0, \pi]$; $\tan^{-1} \in (-\pi/2, \pi/2)$

Tip: Always check dimensional consistency of your answer. If the dimensions don't match, the formula is wrong.

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61. Mathematical Reasoning

Logic

- Negation: $\sim p$
- Conjunction: $p \wedge q$ (AND); Disjunction: $p \vee q$ (OR)
- Conditional: $p \Rightarrow q$ (if p then q)
- Contrapositive: $\sim q \Rightarrow \sim p$ (logically equivalent to $p \Rightarrow q$)
- Converse: $q \Rightarrow p$; Inverse: $\sim p \Rightarrow \sim q$
- Biconditional: $p \Leftrightarrow q$ (p iff q)
- Tautology: always true; Contradiction: always false
- Negation of $\forall x, P(x)$: $\exists x, \sim P(x)$
- Negation of $\exists x, P(x)$: $\forall x, \sim P(x)$



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